**Presentation 5**
• Links to presentation(s) and code(s) on GitHub
Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallOutcomesFinal_9Dec_EDA2.pdf

Link to Recording of Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Dec9_EDASimple2.txt

Codes:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/CallOutcomeTagging_9Dec_Vivienne.ipynb (I did not directly code for this assignment, but I did help write out some of the dictionaries for tagging for Vivienne to run on her machine)

**• What did you do?**
I wrote some of the dictionary tagging for the outcome types and subtypes for Vivienne to compile together in her continuation of Krish's outcome tagging code.

**• How does it help the project?**
This work helped categorize outcomes for our presentation and provided a top-down view of how call volumes and outcomes were distributed across categories. By taking on part of the task, I helped streamline the workflow and speed up the process while Vivienne and Sabrina developed the Python functions to assign outcomes and subtypes to all contact session IDs.

**• Issues faced (if any)**
We ran into some challenges figuring out how to transfer our previous work in R into Krish's code in Python.

**• Attempts to resolve issues (if any)**
We ultimately decided it would be most efficient to keep everything in a consistent dictionary format in Python so we could easily map outcomes to contact session IDs.

**• Issues resolved (if any)**
N/A

**• Next steps**
N/A (end of quarter)

• References (mention if you build in someone else's work)
We built from Krish code:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/caller_type_tag_1Dec_Krish.ipynb

**Presentation 4**
• Links to presentation(s) and code(s) on GitHub
Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/OutcomeTagging_Nov18_EDASimple2.pdf

Link to Recording of Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Nov18_EDASimple2.txt

Codes:
Task on Metric 1:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/OutcomeVisualizationBySubcategory_Nov18_NikoleMontero.R (before running this code, please run the outcome tagging code, which was worked together as a group: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/OutcomeTagging_Nov17_CarolineCorr.R )

**• What did you do?**
Caroline, Hailey, and I split up the last activities in the Excel sheet and wrote cases when statements to map all of their outcomes/sub-types. Then, I computed the distribution of outcomes across all calls and then performed a second aggregation to determine how each outcome type breaks down into its underlying sub-categories. Finally, I produced a set of visualizations, including a stacked bar chart and a faceted breakdown, showing both the proportion of Positive, Neutral, and Negative calls and how each of those categories splits into its major sub-types. These visualizations highlight where the majority of call failures occur (e.g., abandonment, closed queues, after-hours calls) and where successful outcomes tend to originate (voicemail or agent reach).

**• How does it help the project?**
It helps identify why calls end the way they do, pinpointing major pain points in the intake system. It also highlights which failure modes most affect callers (e.g., abandonment, closed queues, after-hours), and it supports prioritizing system improvements by showing where the largest drop-offs occur.

**• Issues faced (if any)**
N/A

**• Attempts to resolve issues (if any)**
N/A

**• Issues resolved (if any)**
N/A

**• Next steps**
Going forward, we should clean up the remaining unclear or edge-case activity patterns, so every call is consistently tagged. Talking with Cynthia, Mark, or the EDA 1 team may help us clarify how certain unusual call flows should be classified. We also need to take a closer look at how outcomes are assigned when callers move across multiple menus. Some calls that eventually reach an agent or voicemail might be getting counted too early, which can underestimate positive outcomes.

• References (mention if you build in someone else's work)
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/OutcomeTagging_Nov17_CarolineCorr.R (this code must be run before my visualization code)

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Clarifications/CAR_data_Activity_name_tagging_10Nov.xlsx

For data import:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CAR_data_import_R_CarolineCorr.R

**Presentation 3**
• Links to presentation(s) and code(s) on GitHub
Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallMenuOutcomes_4Nov_EDATeam2.pdf

Link to Recording of Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/VideoLink_Nov5_EDASimple2.txt

Codes:
For importing datasets:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CarDataImport_Oct21st_Nikole.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/AllCallsDataImport_Oct21st_Nikole.ipynb

Completing task on Employment Menu:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/EmploymentMenu_4Nov_NikoleMontero.R

**• What did you do?**
I filtered and analyzed the CAR dataset for Employment Menu calls after March 16, 2025. Then, I classified those calls into positive, neutral, and negative outcomes based on termination reasons. Finally, I created a visual plot (bar charts) to display the overall call distribution.

**• How does it help the project?**
This analysis helps identify data completeness issues within the Employment flow. It provides a baseline understanding of how call outcomes are recorded post–system update. It also shows that maybe we need to perform a data integration with All Calls to recover or gather some information on queue-level insights.

**• Issues faced (if any)**
Most Employment calls had missing (N/A) queue names or in activity_name. Termination reasons were often missing, leaving most of the calls classified as unidentified, which was considered in the negative category.

**• Attempts to resolve issues (if any)**
I explored filtering by both flow_name and activity_name to recover possible Employment-related entries. I also checked multiple columns (activity_name, queue_name) and time ranges to verify where queue details were stored.

**• Issues resolved (if any)**
I verified that queue-level details are not recorded in CAR after March 16, 2025. So, this probably is a data availability issue, not a coding or filtering error.

**• Next steps**
I would like to request some clarification on the missing queue and termination fields within the Employment Menu Calls.

**Presentation 2**
• Links to presentation(s) and code(s) on GitHub
Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallCategorization_Oct23_Group2.pdf

Codes:
For importing datasets:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CarDataImport_Oct21st_Nikole.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/AllCallsDataImport_Oct21st_Nikole.ipynb

Objective 3:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/IdentifyingPreTenantCalls_23Oct_Nikole.R

**• What did you do?**
I analyzed the Pre-Tenant Menu interactions within the CAR dataset to determine how many callers reached this menu and how many ended the call afterward. I filtered and categorized the call sessions by whether they continued to the Tenant Menu or stopped after the Pre-Tenant Menu, and visualized the distribution of the caller behavior.

**• How does it help the project?**
This analysis helps evaluate whether the Pre-Tenant Menu message is effective in reducing unnecessary tenant calls. If most callers continue despite hearing the message, it suggests the menu adds little value and could be shortened or removed to improve caller flow and reduce wait times.

**• Issues faced (if any)**
Some callers had multiple PreTenantMenu entries under the same session ID, often only seconds apart. This made it difficult to determine whether they were re-listening to the same message or restarting the call.

**• Attempts to resolve issues (if any)**
I addressed this by keeping only distinct session IDs and examining timestamp differences to confirm that duplicates reflected repeated interactions rather than unique calls.

**• Issues resolved (if any)**
I resolved repetition in PreTenantMenu entries by filtering for distinct contact session IDs, ensuring that each call session was counted only once in the analysis.

**• Next steps**
Verify operational definitions of "abandonment" by checking the call flow or verbiage script. Suggest to the Legal Aid team to remove or modify their PreTenant Menu and monitor the call patterns again to assess behavioral changes.

**Presentation 1**
• Links to presentation(s) and code(s) on GitHub
Presentation:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Data%20integrity%20checks/CARAllCallsIntegrity_7Oct_EDATeam2.pdf

Codes:
Importing datasets:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CarDataImport_Oct21st_Nikole.R

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/AllCallsDataImport_Oct21st_Nikole.ipynb

My plot:
https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Time%20Trends%20analysis/Activity%20Pattern_Oct14_Nikole.R

**• What did you do?**
I helped organize and clean the datasets, collaborating with my team to determine the most effective way to align *All Calls* and *CAR* data for comparison. I created an additional plot showing the hourly share of inbound calls to analyze discrepancies between both datasets.

**• How does it help the project?**
This work enhances data consistency and visualization, allowing the team to identify mismatches between *CAR* and *All Calls* datasets more effectively, particularly around inbound call patterns and time alignment.

**• Issues faced (if any)**
The main issue was a mismatch in time variables—specifically, *report_time* from *All Calls* was stored in UTC, while *activity_start_timestamp* in *CAR* appeared to be in Central Time. This caused the hourly distribution of calls to appear misaligned between the two datasets. In addition, the *All Calls* dataset initially included both inbound and outbound calls, leading to further discrepancies since calls from the six key phone lines were not isolated.

**• Attempts to resolve issues (if any)**
I converted *report_time* from UTC to Central Time (America/Chicago) to align both datasets' timestamps. I also refined the filtering process to include only inbound calls from the six phone lines defined in the code, which correspond to those listed in the *AllCallsData_CARData_overlap.txt* clarification file. By restricting the dataset to calls with direction "TERMINATING" and vendor "CallTower," the data now focuses exclusively on relevant inbound calls.

**• Issues resolved (if any)**
After converting the timestamps and filtering for inbound calls, the hourly distributions between datasets became consistent. This confirmed that the earlier mismatch was caused by time zone differences and inclusion of outbound calls. The corrections successfully aligned the *All Calls* and *CAR* datasets for accurate comparison, and when the Hourly Share of Inbound Calls plot was generated, both datasets were successfully matched and aligned.

• Next steps

Since the *All Calls* and *CAR* datasets now align closely after filtering and time conversion, my next step will be to interpret the call patterns more deeply. For example, identifying peak hours of inbound activity and exploring whether certain time periods correspond to higher call volumes or performance differences across the two datasets.